Henry Edwards

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Robotics Software Engineer

Robotics Software Engineer at **DefendTex**, leading **flight-dynamics modelling** and **control-system design** across a **rocket-propelled counter-UAS** platform and a **GNSS-denied drone swarm**. Develop autonomy software in C++ and ROS 2 on NVIDIA Jetson and i.MX 8 embedded Linux platforms, with work spanning Kalman-filter-based state estimation, guidance, SLAM, motion planning, and distributed multi-vehicle control.

Previously founded **SparSafe Helmets**, a sports-safety startup focused on reducing brain-injury risk in boxing. Through the **Melbourne Accelerator Program**, led stakeholder discovery and iterative product development, using feedback from athletes and coaches to define requirements and validate design decisions.

Recently started an independent simulation project exploring **natural-language task execution** and **long-horizon coordination** for **heterogeneous construction-robot fleets**.

Professional Experience

Robotics Software Engineer, DefendTex **Dec 2025 – present**

Technical Lead, Flight Dynamics and Control Systems - Guided RPG40G:

- Developed an in-house simulation toolkit in three weeks to identify causes of early flight stability issues.
- Proposed and implemented an approximately $20 solution that improved accuracy 5X and restored viability of a critically at-risk project.

Subsystem Owner, Terminal Guidance and UWB Swarm Control:

- Led terminal guidance and swarm-control development for a GNSS-denied drone swarm
- Implemented a ROS2 based optical proportional navigation system capable of tracking stationary and moving targets under high disturbance conditions
- Implemented UWB based, bandwidth limited relative swarm position control.

Distributed Systems Communication:

- Designed and built a MAVlink Router in Go for swarming applications, enabling drone-to-drone communication across bandwidth limited radio frequencies.

Founder, SparSafe Helmets **Jan 2024 – Dec 2025**

- Built and released a high-fidelity multiplayer VR boxing simulation in Unity, using user and stakeholder feedback to test assumptions, while participating in the **Melbourne Accelerator Program**.
- Pivoted from VR-first to a physical helmet design, achieving a **50% reduction in key rotational-acceleration metrics**.

Robotics and Control Systems Projects

Construction-Robot Fleet Coordination (early stage, recently started)

- Exploring how heterogeneous robot fleets could execute natural-language instructions and coordinate long-horizon construction tasks.
- Approaches under exploration include vision-language-action (VLA) policies, agents-as-policy, world-action models and Dreamer-style world-model learning
- Drawing on Voyager, Dreamer and π0 to inform the simulation testbed and policy design.

Deep Reinforcement Learning for Robotic Control

- Implemented reinforcement-learning agents from scratch using tabular Q-learning, DQN, vanilla policy gradient (VPG) and Soft Actor-Critic (SAC)
- Applied agents to gridworld navigation, inverted-pendulum balancing and HalfCheetah locomotion in Gymnasium using Python, PyTorch, NumPy and CUDA

Technical Skills

- **Languages:** C++, C, Rust, Go, MATLAB, Java
- **Robotics & AI:** ROS, ROS2, IsaacSim, PyTorch, TensorFlow, OpenCV, SLAM, MuJoCo
- **Software & Control:** Simulink, Unity, Git, TCP/IP, Embedded Systems Design, Model Predictive Control, LQR
- **CAD & Hardware:** SolidWorks, Fusion 360, NVIDIA Jetson, Raspberry Pico, i.Mx8 chipsets.

Education

Master of Mechatronics Engineering Jul 2023 – Nov 2025
The University of Melbourne

- Relevant Coursework: Advanced Control Systems, Autonomous Systems, Robotics, and Mechatronics Systems Design

Bachelor of Science Mar 2018 - Nov 2021
The University of Melbourne

- Major in Mechanical Engineering

Volunteer and Extra-curricular Activity

- Freeride World Qualifier 2* Athlete (Competitive Downhill Skiing)
- Volunteer at Santa Teresa Lifeguards in Costa Rica